# Item 4 briefing — Selection of the depth-retrieval criterion

Comparison of three depth-retrieval methods | HydroLight independent set of 288 cases | 2026-09-03

**Problem**

The existing algorithm estimates a single common depth z from the $R_{rs}$ reconstruction error over four bands (443, 490, 560 and 665 nm) by scalar minimization of a cost function. That multi-band procedure was not stated clearly enough in earlier reports, so the algorithm could be read as if a different depth were obtained at each wavelength, with no stated rule for choosing a final depth. It was therefore proposed that a depth $H(\lambda_k)$ be computed independently in each band, that $R_{rs}$ be reconstructed from each of those depths, and that candidate solutions be ranked by the mean reconstruction residual.

**Objective**

The aim was to distinguish the existing four-band scalar-minimization scheme from the proposed per-band depth and mean-residual scheme, and to compare their depth-retrieval accuracy so that a more stable inversion criterion could be selected. The comparison also included the use of all four bands versus the three bands of relatively higher reliability for depth retrieval.

**Procedure**

The same validation set was inverted with three depth-retrieval methods. The difference is only what is minimized, as written in Table 1. Method 1 computes a depth in each band and ranks candidates by the mean residual $\bar{\varepsilon}$; Method 2 finds one common depth z* that minimizes the sum of squared errors over four bands; Method 3 uses the same cost as Method 2 but on three bands only.

The validation set comprises 288 HydroLight cases (6 bottom types × 6 depths × 2 solar zenith angles × 2 chlorophyll levels × 2 suspended-matter levels). These 288 cases do not overlap the 972 cases used to fit the coefficients of Eqs. 2 and 4. All three methods were given the true bottom reflectance; absorption a and backscattering $b_b$ were taken from QAA. Retrieved depths were compared with true depths using RMSE and MAE, separately for Eqs. 1–4.

**Table 1. The three methods — only the quantity being minimized differs**

| Method | In plain words | Equation |
|---|---|---|
| Method 1 (proposed) | (1) At 443, 490 and 560 nm, find a depth $H_k$ that matches that band alone.<br>(2) Reconstruct all four bands with each $H_k$ and take the mean residual $\varepsilon_k$.<br>(3) Use the $H_k$ whose $\varepsilon_k$ is smallest. | $H_k = H(\lambda_k)$, k = 1,…,N<br>$\varepsilon_k = (1/M)\ \Sigma_j\ \lvert R_{rs}^{input}(\lambda_j) - R_{rs}^{sim}(\lambda_j; H_k) \rvert$<br>Cost = $\bar{\varepsilon} = (1/N)\ \Sigma_k\ \varepsilon_k$ |
| Method 2 (existing 4-band) | Place a single depth in the forward model, sum the squared errors over four bands, and take the common depth z* that makes that sum smallest. No per-band depth is chosen. | $z^* = \arg\min_z \Sigma_\lambda\ [\, R_{rs}^{input}(\lambda) - R_{rs}^{sim}(\lambda; z)\, ]^2$<br>λ = 443, 490, 560, 665 nm |
| Method 3 (reliable 3-band) | Same equation as Method 2, but the cost uses only 443, 490 and 560 nm. | $z^* = \arg\min_z \Sigma_\lambda\ [\, R_{rs}^{input}(\lambda) - R_{rs}^{sim}(\lambda; z)\, ]^2$<br>λ = 443, 490, 560 nm |

Notation. N = number of trusted bands (3); M = number of bands compared (4). $R_{rs}^{input}$ is the observed (simulated) reflectance; $R_{rs}^{sim}(\lambda; H)$ is the value reconstructed at depth H. The Method-1 cost $\bar{\varepsilon}$ scores a candidate; the representative depth is the $H_k$ with the smallest $\varepsilon_k$.

## Why three bands were treated as the reliable set

When the true depth is substituted into the forward model, as in Item 1, the relative $R_{rs}$ error is 2–8% at 443, 490 and 560 nm, but 13–21% at 665 nm—more than three times larger (Figure 1). $R_{rs}$ itself is small at 665 nm, so a given absolute mismatch appears large as a relative error. For that reason 665 nm was treated as the less reliable band, and Methods 1 and 3 used only 443, 490 and 560 nm.

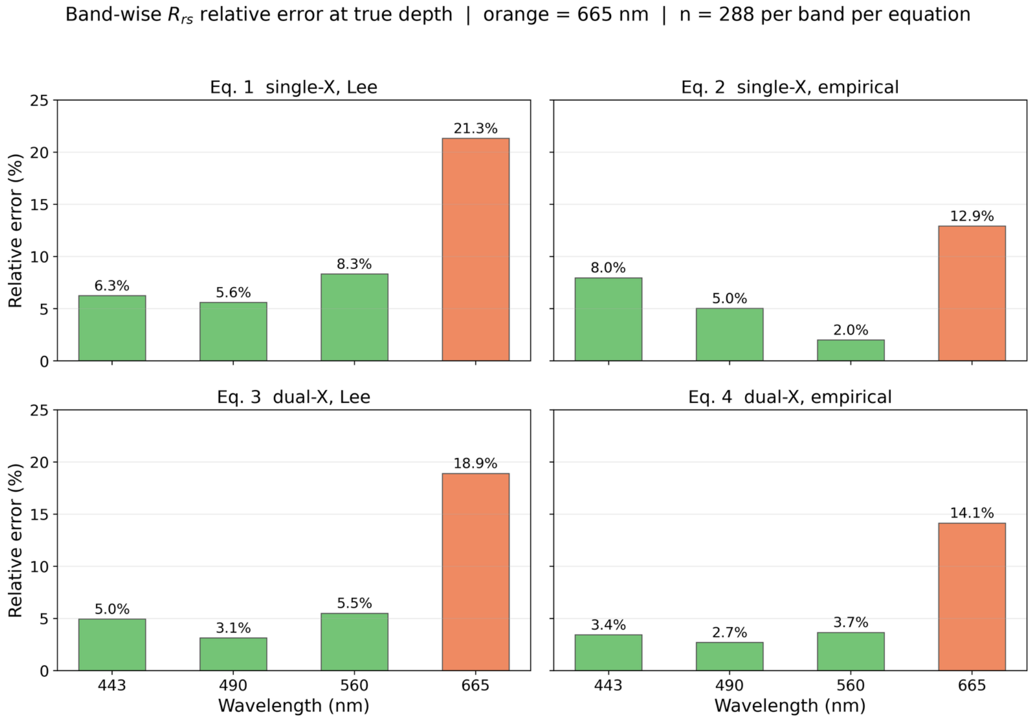


Figure 1. Band-wise relative $R_{rs}$ error when the true depth is substituted (Eqs. 1–4; n = 288 per band). Green = 443, 490 and 560 nm; orange = 665 nm. Relative error at 665 nm alone is 13–21%, which is why Methods 1 and 3 omit that band.

# Results

## 1) Depth accuracy — Method 2 is the most accurate

Depths for the 288 cases were retrieved by the three methods and compared with the true depths (Figure 2, Table 2). Method 2 was the most accurate (RMSE 0.39 m), followed by Method 3 (0.42 m) and Method 1 (0.50 m). Under Method 1 the three per-band depths differed by 1–2 m on average even for the same case, and depths from 443 and 490 nm were highly inaccurate. The band with the smallest residual was 560 nm in 62–72% of cases, so Method 1 effectively reduced to a 560 nm retrieval. Method 3, by dropping 665 nm, lost information that is sensitive to shallow depths and was therefore worse than Method 2.

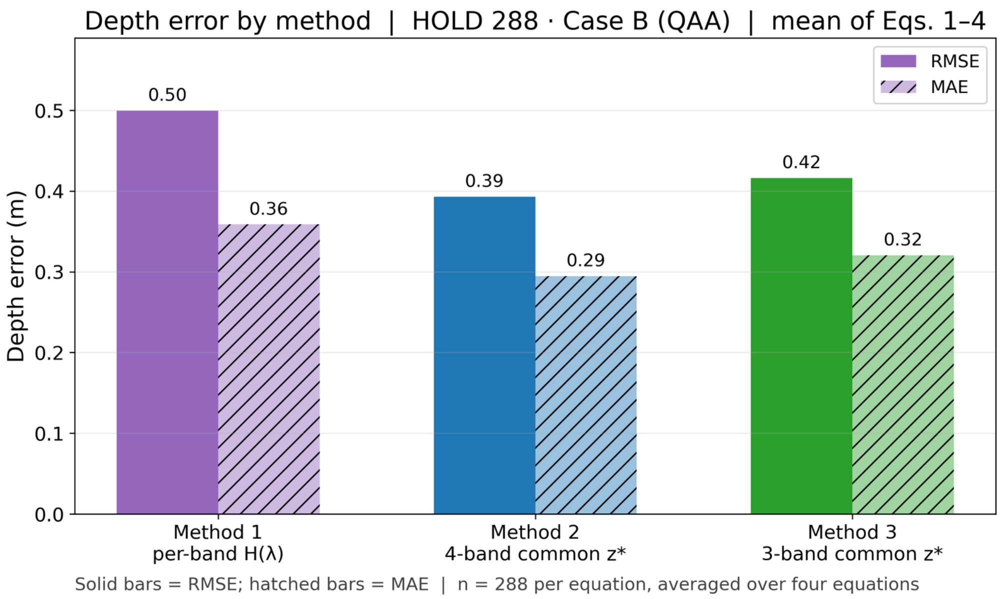


Figure 2. Depth error of the three methods (mean of Eqs. 1–4). Solid bars are RMSE and hatched bars are MAE; numbers above the bars are the plotted values. Method 2 (blue) is the smallest.

**Table 2. Depth error by method (288 cases; mean of Eqs. 1–4)**

| Method | RMSE (m) | MAE (m) | MAPE (%) |
|---|---|---|---|
| Method 1 — selected per-band depth | 0.50 | 0.36 | 15.4 |
| Method 2 — four-band common depth | 0.39 | 0.29 | 11.9 |
| Method 3 — three-band common depth | 0.42 | 0.32 | 14.8 |

Note. The highlighted row is the selected Method 2. The ranking was unchanged when HydroLight true a and $b_b$ were used, and when the comparison was repeated on the 972-case set.

## 2) Method 2 depth retrieval (Eqs. 1–4)

Retrieved depths from Method 2 are compared with true depths in Figure 3. All four equations follow the 1:1 line over 0.5–5 m. Equation-wise errors are given in Figure 4 and Table 3. Eq. 3 was the most accurate (RMSE 0.33 m); Eqs. 4 (0.35 m) and 2 (0.36 m) were similar, and Eq. 1 was the poorest (0.53 m). Equations with the original Lee coefficients (1 and 3) tended to overestimate depth; those with empirical coefficients (2 and 4) tended to underestimate it.

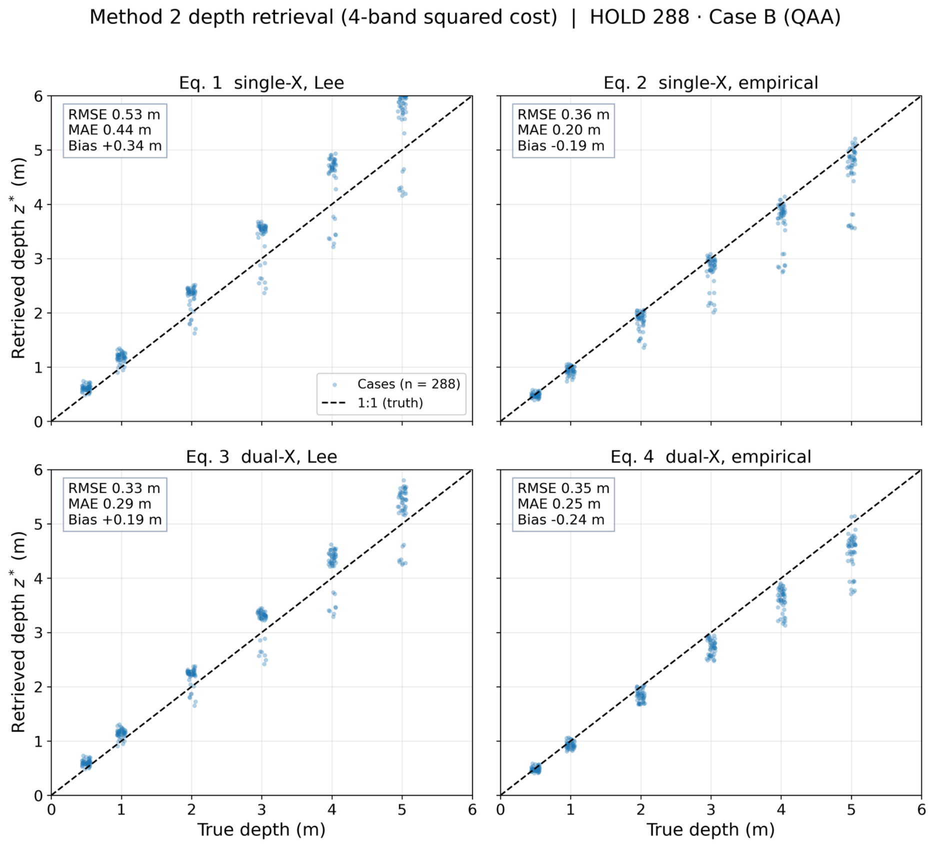


Figure 3. Method 2 depth retrieval (Eqs. 1–4; n = 288 each). The x-axis is true depth and the y-axis is retrieved depth; each point is one case. Closer to the dashed 1:1 line is more accurate. The inset box gives RMSE, MAE and bias for that equation.

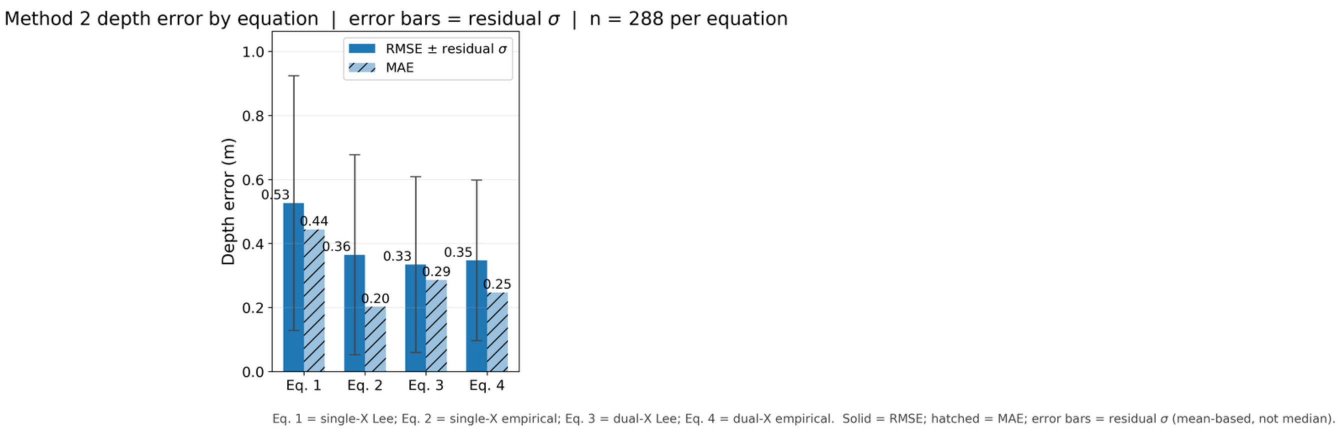


Figure 4. Method 2 depth error by equation (n = 288 each). Solid bars are RMSE and hatched bars are MAE; both are mean-based statistics (not medians). Vertical error bars on RMSE are the standard deviation of the residual (retrieved minus true). Eq. 3 is the smallest and Eq. 1 is the largest.

**Table 3. Method 2 depth error by equation (288 cases per equation)**

| Equation | RMSE (m) | MAE (m) | Bias (m) | MAPE (%) |
|---|---|---|---|---|
| 1 single-X, Lee coefficients | 0.53 | 0.44 | +0.34 | 18.1 |
| 2 single-X, empirical coefficients | 0.36 | 0.20 | -0.19 | 7.8 |
| 3 dual-X, Lee coefficients | 0.33 | 0.29 | +0.19 | 12.9 |
| 4 dual-X, empirical coefficients | 0.35 | 0.25 | -0.24 | 8.9 |

Note. Positive bias means the retrieval is too deep; negative bias means it is too shallow.

## 3) $R_{rs}$ reconstructed from the retrieved depth (Eqs. 1–4)

The retrieved depth z* was substituted back into the forward equation and the resulting $R_{rs}$ was compared with HydroLight. In Figure 5 the model (blue) and HydroLight (red) means and medians nearly overlap for all four equations, showing that the spectrum is well reproduced at the retrieved depth. Band-wise RMSE (Figure 6) is larger at 560 nm, where the signal is strongest, and at 443 nm, where QAA error is larger, and is smallest at 490 nm. The error at 665 nm is comparable to that at 490 nm, so including 665 nm in the cost is not a problem.

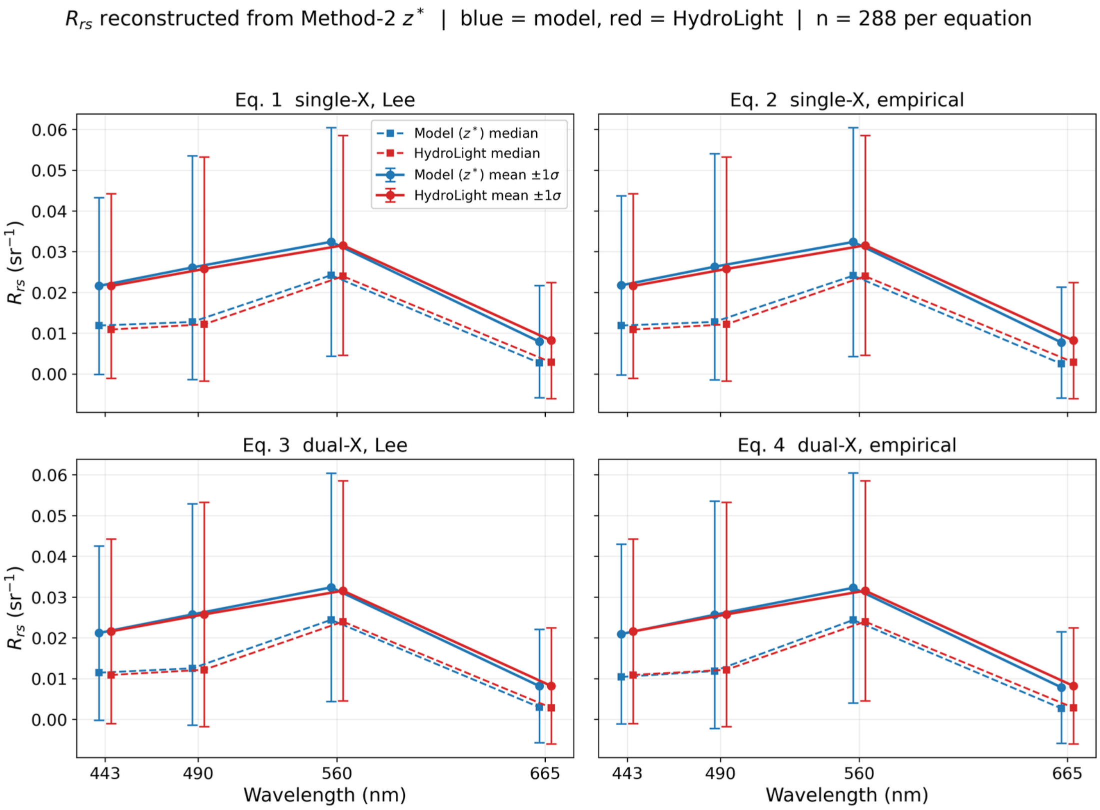


Figure 5. $R_{rs}$ reconstructed from Method-2 depths (Eqs. 1–4; n = 288 each). Blue = model; red = HydroLight. Solid lines are means with vertical bars of ±1σ; dashed lines are medians. Closer overlap of the blue and red lines indicates better reconstruction.

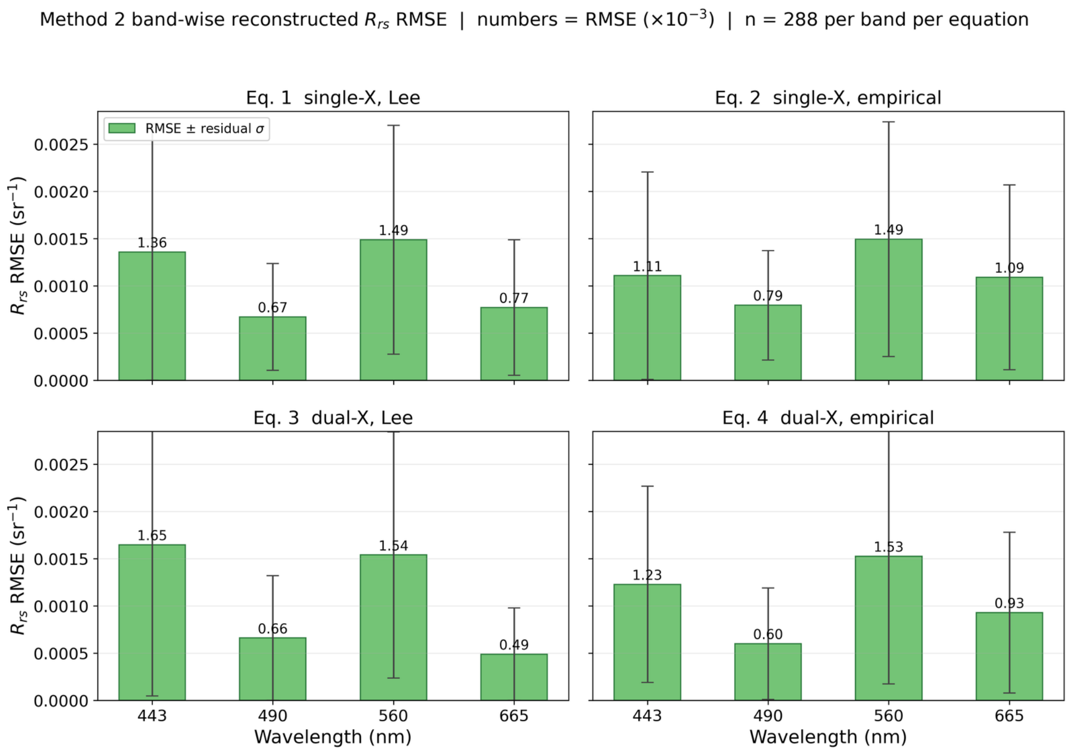


Figure 6. Method 2 band-wise $R_{rs}$ RMSE (Eqs. 1–4; n = 288 per band). Bars are RMSE; numbers above the bars are RMSE (×10$^{-3}$); vertical error bars are the standard deviation of the residual.

## Conclusion

Conclusion

When depth is obtained separately in each band, the three depths for the same case differ by 1–2 m on average and by as much as 3–5 m. Depths from 443 and 490 nm are particularly inaccurate, so Method 1 effectively selects the 560 nm depth and is less accurate than Method 2 (RMSE 0.50 m versus 0.39 m). Choosing one band as a representative depth is therefore not justified.

The depth-retrieval criterion is therefore Method 2: a single common depth z* obtained by scalar minimization of a reconstruction-error cost over four bands (443, 490, 560 and 665 nm). There is no step that selects a band, and both the depth error and the $R_{rs}$ reconstruction error are the smallest of the three methods. Relative error at 665 nm is large, but that band still carries information on shallow-depth change; omitting it (Method 3) degrades accuracy, so all four bands are retained. Eqs. 3 and 4 are used, with empirical-coefficient Eq. 4 preferred. These rankings were obtained on the 288 cases that were not used to fit coefficients, and they were unchanged with true a and $b_b$ and on the 972-case set.

The proposed mean of per-band reconstruction residuals is not used to choose the depth. It is used in the next step to rank seafloor candidates: Method 2 supplies z* for each candidate, and the mean reconstruction residual is then compared among candidates.

Limitation. This comparison used the true bottom reflectance. A comparison in which bottom type and mixing fractions are also unknown is left to Item 5 (mixing-fraction inversion).